

Why Math matters for Deep Learning :

- Understand how models work
- Debug when something is not working
- Tune parameters with intent
- Build new architectures

img $\rightarrow$ $\log(0.7)$
cat (0.2)
horse (0.1)

The 3 Pillars:

Linear Algebra :

- Store + move data
- Scalars, vectors, matrices and tensors

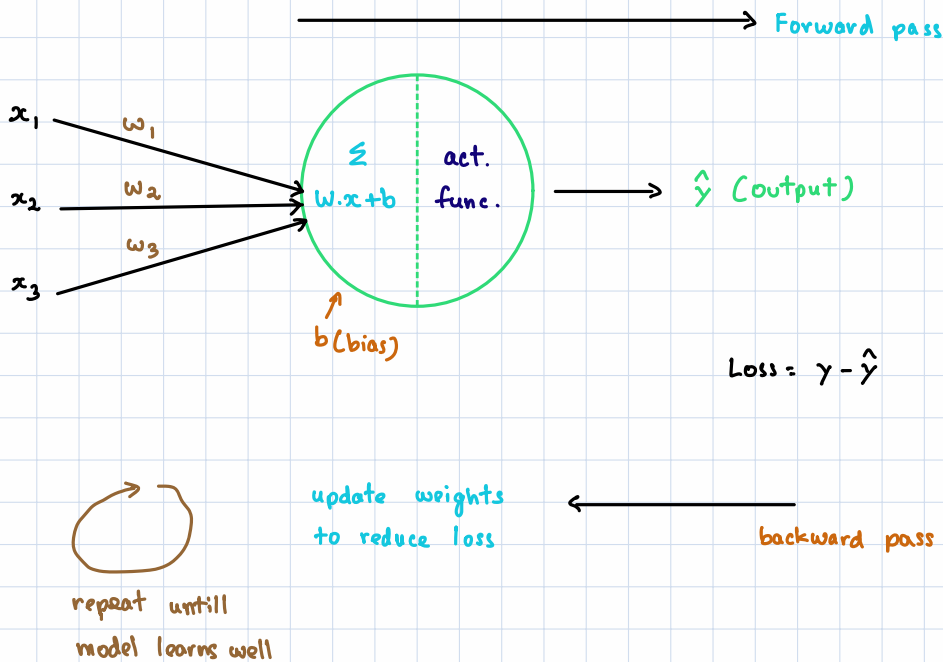
Calculus :

- tells the model how to improve
- gradients, derivatives, chain rule

Probability :

- handling uncertainty
- distribution, mean, variance

Math in Action: One Neuron:



$w.x + b$ $\rightarrow$ Linear Algebra $\rightarrow$ Matrix multiplication

Backward pass $\rightarrow$ Calculus $\rightarrow$ Gradient Descent

$\hat{y}$ (pred. prob) $\rightarrow$ Probability